

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Defined and executed an **AI strategy** for core Bitcoin mining platforms (MOS/MDK), focusing on operational efficiency and measurable business outcomes by translating site-level challenges into **data and optimization problems**.
- Designed a **standardized data baseline and schema** for mining-site telemetry and events, ensuring data was structured, normalized, and accessible for **AI/ML use cases**.
- Built and deployed **ML/AI and optimization models** including **predictive failure analysis** and **energy-efficiency insights**, integrating them into platform workflows and APIs.
- Collaborated with backend engineers to evolve the **data pipeline and integration standards**, ensuring high-quality **data modelling** for AI-driven solutions.
- Worked with **site-operations teams** to validate model results and gather feedback, ensuring AI-driven recommendations were practical and actionable in production.
- Developed **Python scripts** to automate data extraction and transformation from diverse sources, supporting the creation of structured datasets for model training and analysis.
- Containerized **Java Spring Boot microservices** with **Docker**, deploying them to an **AWS** environment and managing CI/CD pipelines via **Jenkins**.

SKILLS

AI/ML – Machine Learning, AI Strategy, Optimization Models, Predictive Failure, Anomaly Detection, Energy Efficiency, Data Modelling, Mathematical Optimization

Data Engineering – Data Pipeline, Data Architecture, Data Schema, Data Baseline, ETL, Data Warehousing, SQL, PostgreSQL, MySQL, Oracle

Programming Languages – Python, Java, SQL, R

Backend Development – Spring Boot, REST APIs, Microservices, FastAPI, Flask

Cloud & DevOps – AWS, EC2, S3, Lambda, CloudWatch, Docker, Jenkins, Git

Analytics & BI – Power BI, Tableau, DAX, Power Query, Pandas, NumPy, Scikit-learn

Operations & Systems – Linux, Shell Scripting, Splunk, Dynatrace, Production Support, Incident Response

PROJECTS

Bitcoin Mining Operations Optimization Platform (HCL Technologies)

- Designed and implemented a **data architecture** for mining site telemetry, integrating data from various sensors and operational logs into a unified **PostgreSQL** database.
- Developed **Python** scripts to process and clean raw telemetry data, transforming it into feature sets for predictive maintenance models, reducing data processing time by **~40%**.
- Built **dashboards in Power BI** to visualize key operational metrics and model outputs, enabling site managers to track performance and identify areas for improvement.

Predictive Maintenance Model for Industrial Equipment (MSc Applied Project)

- Engineered a **feature pipeline** using **Python** and **Pandas** to extract time-series features from sensor data for industrial machinery, processing over **1.5 million data points**.
- Trained a **scikit-learn** classification model (Random Forest) to predict equipment failure within a 7-day window, achieving an **AUC of 0.89** on a held-out test set.
- Deployed the trained model as a simple **REST API** using **FastAPI** for integration into a simulated operational monitoring system.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)